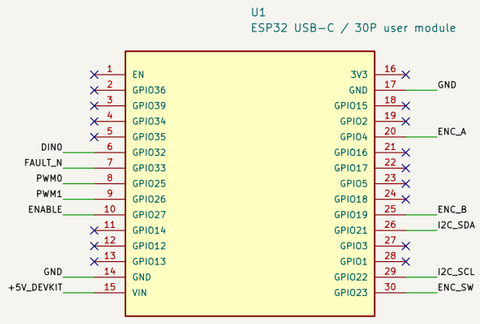
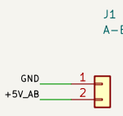
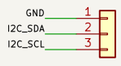
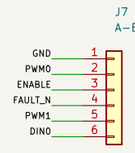
A1 — HOST AND COMMON INTERFACE / LOGIC 3.3 V



U1 is a removable DevKit, not a bare ESP32.
Common 30P pinout; row spacing25.40mm confirmed. Two Harwin M20-7821546 sockets.



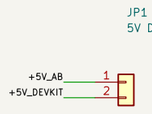




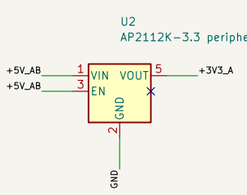
J1 POWER: 1 GND / 2 +5V. J6 I2C: 1 GND / 2 SDA / 3 SCL.
J7 CTRL: 1 GND / 2 PWM0 / 3 ENABLE / 4 FAULT_N / 5 PWM1 / 6 DINO.
Straight-through cables, JST-XH 2.50 mm. NO 24 V, heater current or 3V3 rail.



A2 — LOCAL POWER / LCD 3.3 V AS CURRENT PROTOTYPE

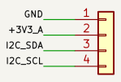








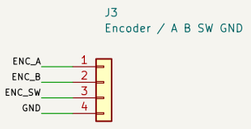




LCD header order is a proposed cable order.
Check actual backpack pin order, 3.3 V operation
and backlight current before PCB layout.

JP1 fitted in normal use. Do not parallel external 5 V and USB VBUS.
USB maintenance: B off, disconnect J1/J6/J7, remove JP1; no peripheral operation in this mode.
Powered serial debugging requires a verified VBUS-isolated USB connection. No automatic power mux.

A3 — USER INPUT / I2C PERIPHERALS



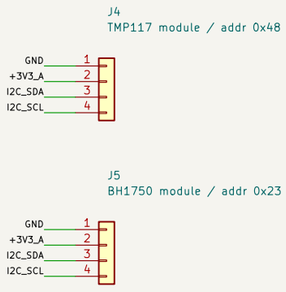






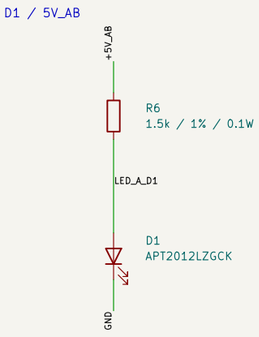


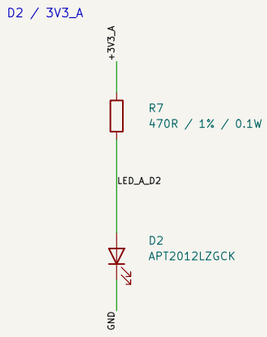




Only R4/R5 are the main I2C pull-ups. Remove or account for pull-ups on all breakout modules and LCD.
DevKit 3V3 is not joined to +3V3_A. U2 supplies peripherals only; check current and thermal budget.
I2C cable A-B: short internal harness. ADC on B = 0x49, TMP117 on A = 0x48.

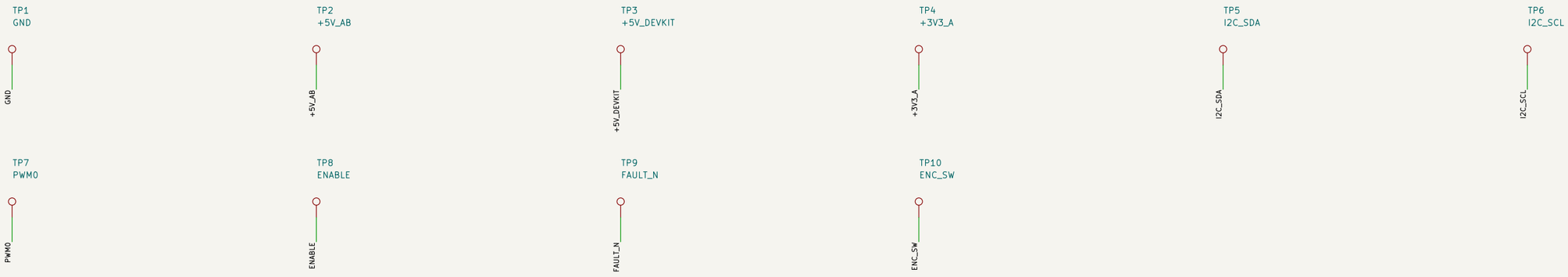
DIAGNOSTICS — RAIL PRESENCE AND HEATER GATE DRIVE





Green LEDs show rail presence only, not regulated voltage accuracy or ESP32 execution.
HEAT (B/D8): 5 V driver output after PWM AND ENABLE; brightness follows duty, NOT measured heater current.
NTC on B/J4: 1 = +3V3_B, 2 = NTC_RAW; passive two-wire 15k B3920 probe, either wire orientation.

BENCH TEST PADS — BACK SIDE B.Cu — COPPER ONLY, DO NOT POPULATE



PROTOTYPE — ENGINEERING PREPARED — ASSEMBLER APPROVAL REQUIRED

Laboratory hardware platform

Sheet: /
File: pcb_a_display.kicad_sch

Title: PCB A — Display and ESP32 carrier

Size: User	Date: 2026-11-05	**Rev: 0.4.1**
KiCad E.D.A. 10.0.6		Id: 1/1